

SPACE LAUNCH REPORT: FALCON 9 V1.2

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Falcon 9 v1.2

SpaceX Reusable Booster Workhorse

F9-21, the First Falcon 9 v1.2

During early 2015, SpaceX announced Falcon 9 upgrades from its version 1.1 (v1.1) that would allow first stage landings during geosynchronous transfer missions. The upgrades would include a 15% increase in thrust, the use of "deep cryogenic", or "densified", liquid oxygen, a 10% second stage tank stretch, a longer second stage engine nozzle, and a stretched interstage. The upgraded rocket would provide about a 30% increase in performance. For a while SpaceX called it "Falcon 9 Full Thrust", but by late 2015 or early 2016 the company had filed the name "Falcon 9 v1.2" with the FAA.

Falcon 9 v1.2 would lift off on 694 tonnes of sea level thrust rather than Falcon 9 v1.1's 600 tonnes. Second stage Merlin 1D Vacuum thrust would increase to at least 95 tonnes force. The longer second stage engine nozzle would improve specific impulse. Testing of upgraded Merlin engines was completed at McGregor, Texas during the summer. Meanwhile, the first v1.2 first stage departed the SpaceX

Hawthorne, California factory at the end of August and arrived at McGregor a few days later.

On September 8, 2015, the stage was erected at the new ground-level test stand, its first use. The stand, completed in 2013, used a below-grade flame trench to reduce noise imposed on neighboring communities during hot fire tests. On September 21, the stage performed a 15 second test firing. A full duration test was performed on November 19, 2015. The second stage was tested on November 22.

The first stage arrived at Cape Canaveral on November 20, 2015. On December 18, the first stage, topped by the second stage and integrated Orbcomm G2 payload, completed a brief static firing at Space Launch Complex (SLC) 40 after two days of scrubbed attempts that appeared to be ground-system related, as the test shook down new super-chilling equipment at the pad.

SLC 40 at Cape Canaveral and SLC 4 East at Vandenberg AFB gained erector transporter upgrades and new propellant densification plants. Meanwhile, Launch Complex 39A at the Kennedy Space Center was being modified from its STS configuration for both Falcon Heavy and Falcon 9 v1.2, with a new horizontal processing hangar on the former crawlerway at the base of the pad and a new pair of railroad tracks leading up to the launch pad itself.



First v1.2 Flight and First Successful Booster Landing

The first Falcon 9 v1.2 (and 20th Falcon 9) launch on December 22, 2015 boosted 11 Orbcomm satellites into orbit from SLC 40. For the first time, a Falcon 9 booster landed successfully, boosting back to Cape Canaveral Landing Zone 1 at the former Launch Complex 13. The 69.799 meter (229 foot) tall rocket was about 1.524 meters (5 feet) taller than v1.1. First stage B1019 fired for 2 minutes 20 seconds. The second stage ignited its improved Merlin Vacuum engine 15 seconds later to begin a nearly eight-minute burn to reach a roughly 620 x 660 km x 47 deg orbit. The eleven 172 kg Orbcomm satellites separated in sequence 15 to 20 minutes after liftoff. After spacecraft deployment, the second stage performed a deorbit burn.

Meanwhile, B1019 performed a roughly 30 second boostback burn beginning about 3 minutes 50 seconds after launch and a 20 second reentry burn about 8 minutes after launch, both using three engines. A final, roughly 32 second single-engine landing burn using only the center engine took place just before the landing, about 10 minutes after liftoff. LC 13

had previously supported 51 Atlas missile and Atlas Agena orbital launches from 1958-1978. The site's mobile service tower was demolished in 2005 and its blockhouse in 2012. SpaceX subsequently built an 86 meter (282 foot) diameter landing pad centered on the spot where the original Atlas missile service tower parked during launches.

After the mission the stage was moved to the new Horizontal Integration Facility at LC 39A a couple of days after its landing, where it was photographed and inspected. During week of January 12, the stage returned to SLC 40. A crane was used to lift it onto the stand rather than the usual erector. On January 14, an unannounced static fire attempt was made and aborted after 2-3 seconds when one of the

outer engine's thrust fluctuated. After the test, Elon Musk tweeted that the engine would be borescoped and that it might have ingested something. B1019 subsequently returned to LC 39A and was never refired. Eventually, it was placed on display outside the SpaceX factory in Hawthorne.



Falcon 9 Launches CRS 8, First Stage Lands at Sea

After the second Falcon 9 v1.2 succeeded in boosting SES 9 to supersynchronous transfer orbit on March 4, 2016 but lost its booster during its drone ship landing attempt, it was left to the third Falcon 9 v1.2 (23rd Falcon 9) to accomplish the first successful floating platform landing. The launch orbited the Dragon 10 spacecraft on NASA's CRS 8 International Space Station cargo hauling mission on April 8, 2016. The drone ship landing took place about 300 km northeast of the Cape Canaveral

Space Launch Complex 40 launch site. It was the fifth such attempt. The booster, B1021, performed three-engine boost-back and entry burns, followed by a single-engine landing burn. Landing took place about 8 minutes 35 seconds after liftoff.

Dragon was targeted toward a 200 x 360 km x 51.6 deg orbit. It carried 3,136 kg of cargo, including the 1,413 kg Bigelow Expandable Activity Module (BEAM) experiment in the unpressurized trunk section. At more than 8.6 tonnes, the combined Dragon and cargo mass was the heaviest payload yet launched by a Falcon 9. Dragon arrived at ISS on April 10, 2016.

B1021 had performed a static firing at SLC 40 on April 5. After its successful static testing at McGregor, Texas during February, a ground equipment failure had damaged multiple engines during a non-propulsive test. The engines were repaired or replaced without causing significant delay to the launch schedule.

The drone ship with the first stage returned to Port Canaveral on April 12, 2016. During the day, a crane picked up the stage from the barge and placed it onto a work stand on the dock. After several days of processing which included leg removal, the stage was moved to the Launch Complex 39A HIF on April 19.

"Block 5" Revealed

On April 30, 2016, SpaceX released new performance data for a further improved Falcon 9 v1.2 later identified as "Block 5". Gross mass increased to 564 tonnes and liftoff thrust rose to 775.65 tonnes. Payload rose to 22.8 tonnes to LEO x 28.5 deg, 8.3 tonnes to GTO x 27 deg, and 5.5 tonnes GTO x 27 deg when the first stage was recovered downrange. The cost for a flight with first stage recovery was listed at \$62 million. Dimensionally identical to previous v1.2 versions, Block 5 was designed for Dragon Commercial Crew launches for NASA but would also power unmanned missions. It would begin flying in 2018. Meanwhile, Block 2, 3 and 4 Falcon 9 v1.2 models would introduce incremental improvements.



Falcon 9 and AMOS 6 Destroyed in Pre-Launch Test

A Falcon 9 rocket and its \$200 million AMOS 6 satellite payload were destroyed during a pre-launch propellant loading and hot fire test exercise at Cape Canaveral on September 1, 2016. The test was planned to assure all was ready for a September 3 launch that would have placed 5.5 tonne AMOS 6 in geosynchronous transfer orbit.

Early reports indicated that propellant loading was nearly completed, and the test was about eight minutes away when a powerful explosion destroyed the rocket and satellite at about 9:07 AM Eastern Time. A series of smaller explosions occurred during the following minutes as a fire raged at SLC 40 and a large plume of black smoke drifted across the Florida space center. It was the largest pad explosion in the history of Cape Canaveral/Kennedy Space Center.

A few hours after the explosion, Elon Musk tweeted that the failure appeared to have begun at the second stage liquid oxygen tank. SLC 40 was heavily damaged, knocking it out of service. A day after the failure, SpaceX announced that East Coast launch campaigns would move to Kennedy Space Center Launch Complex 39 Pad A, which at the time was planned to be ready to support operations beginning in November 2016.

The AMOS 6 launch would have been the 29th Falcon launch, and the ninth by a Falcon 9 v1.2 variant. The AMOS 6 first stage, B1028, was test fired at McGregor, Texas on August 5, 2016, and arrived at Cape Canaveral sometime after August 21.

SpaceX subsequently determined that the cause was sudden overpressurization of the second stage liquid oxygen (LOX) tank due to the failure of a composite overwrapped pressure vessel (COPV) containing pressurized helium that was mounted inside the LOX tank. Improper control of subcooled-LOX temperatures may have been involved. Elon Musk suggested that LOX froze within or beneath the composite overwrapping, causing loss of COPV structural integrity. SpaceX performed cryogenic loading tests, with some leading to failure, of small test vessels at its McGregor, Texas test site to confirm the failure mode. The company would change its propellant loading procedures, more than doubling the LOX loading time.



Falcon 9 Returns to Flight

Ending a four-month failure investigation stand-down, Falcon 9 v1.2 returned to service on January 14, 2017, orbiting ten IridiumNEXT satellites from Vandenberg AFB in California. First stage B1029 powered the rocket from SLC 4E at 17:54 UTC to begin a hour-long mission that inserted the 860 kg, Thales Alenia Space-built satellites into 610 x 620 km x 86.4 deg orbits. The launch was the first of seven planned IridiumNext Falcon 9 flights. It was the 29th Falcon 9 launch and the ninth v1.2 variant to fly, not including the lost AMOS 6 launch vehicle. It was the first v1.2 to fly from VAFB.

After a 43 minute, 16 second coast, the second stage restarted for a brief circularization burn about 52 minutes 31 seconds after liftoff to complete the powered phase of the flight. Spacecraft separation began at about the 59 minutes 16 seconds mark, with each satellite separating individually separated by about 1.5 minutes.

B1029 performed boost-back, entry, and landing burns to land on drone ship “Just Read the Instructions”. It was the first successful first stage landing in two West Coast attempts. The first and second stages were test fired at the company's McGregor, Texas test site during late October and early November 2016. The first stage was hot fired at SLC 4E on January 5, 2017, after a scrub the previous day. The payload was not stacked atop the vehicle during the hot firing.



Falcon 9 Debuts from KSC

Falcon 9 v1.2 orbited the CRS-10 Dragon spacecraft with cargo for the ISS from Kennedy Space Center Launch Complex 39 Pad A on February 29, 2017. It was the first Falcon 9 launch from the converted NASA Saturn 5/Space Shuttle launch site. Liftoff took place at 14:39 UTC, following an aborted attempt one day earlier caused by out-of-range readings from the second stage thrust vector control system. CRS-10 (Dragon spacecraft No. 12) carried about 2,490 kg tonnes of cargo, including 1,530 kg inside the pressurized capsule and 960 kg attached to the unpressurized trunk section.

Falcon 9's second stage boosted Dragon into a 51.6 deg low earth orbit, with stage cutoff occurring 9 min 5 sec after liftoff and spacecraft separation taking place one minute later. While the second stage was performing its 393-second-long burn, first stage B1031 did a 180 deg flip and performed 3-engine boostback burn. It flipped again before performing a 3-engine

entry burn and a single engine landing burn that began about 7 min 33 sec after liftoff. The stage landed at Cape Canaveral Landing Zone 1, performing the first daylight landing, and third overall, at the site. The second stage was expected to perform a deorbit burn after spacecraft separation.

At LC 39A the flame trench had been rebuilt and reconfigured, with exhaust now exiting only toward the north. Large "rainbirds" were added to spray water on the launcher during liftoff. It was the 95th launch from LC 39A,



a number that included 12 Saturn 5 and 82 Space Shuttle liftoffs, the most recent by Shuttle Atlantis on July 8, 2011, for the STS-135 mission.

Falcon 9 Reflies First Stage, Orbits SES 10

SpaceX launched a previously flown Falcon 9 first stage for the first time on March 30, 2017. The stage, B1021.2, helped boost the 5,282 kg SES 10 communications satellite to geosynchronous transfer orbit from KSC LC 39A. B1021 had previously flown during the F9-23 CRS-8 mission on April 8, 2016. This time B1021 repeated its previous downrange drone ship landing after performing entry and landing burns. It was the first reflight of a complete orbital-class liquid fueled rocket stage.

After the flight, SpaceX CEO Elon Musk announced that the company had, in another first, directed one of the two payload fairing halves to a landing zone in a test of future payload fairing recovery. The fairing had been equipped with a cold gas thruster system for stabilization during reentry.

After its 2016 flight, B1021 had been shipped back to the SpaceX factory in Hawthorne, California with its engines removed. After the engines were re-installed and other refurbishment work completed, the stage was shipped to the company's McGregor, Texas test site. There, it was test-fired on January 25, 2017, completing what appeared to be a standard test cycle for a Falcon 9 first stage. The new second stage was also test fired in late January or early February. After shipment to LC 39A's Horizontal Integration Facility, the assembled rocket performed a five-second static test at LC 39A on March 27, 2017, with no payload installed.



"Block 5" Inaugural

The first Falcon 9 v1.2 "Block 5", orbited Bangabandhu 1, Bangladesh's first geostationary communications satellite, from KSC LC 39A on May 11, 2018. The 3.72 tonne Thales Alenia Space Spacebus 4000B2 series satellite separated into a geosynchronous transfer orbit about 33 minutes 38 seconds after the 20:14 UTC liftoff, following two burns by the Falcon 9 second stage. The launch followed an automatic abort on May 10 about 1 minute before that planned liftoff.

Block 5 incorporated higher-thrust Merlin 1D engines with turbopump modifications requested by NASA to support crew launches. Block 5 included second stage pressurization changes, made after the AMOS 6 Falcon 9 explosion, that allowed faster, later, denser, chilled kerosene fuel loading. Block 5 also sported new landing legs able to retract after landing without needing to be removed as was prior practice. Other changes appeared in earlier Block designs, including a bolted rather than a welded first stage engine thrust structure, titanium grid fins, and an improved

second stage design. The upgrades allowed for more than that the single first stage reflights so far attained.

First stage number B1046 burned for about 2 min 31 sec, a few seconds shorter than earlier, similar missions, before separating to perform reentry and landing burns to land on the "Of Course I Still Love You" droneship positioned about 630 km downrange. The second stage performed a planned 5 min 43 sec first burn to reach a parking orbit at T+8 min 19 sec. This burn was substantially shorter than earlier, similar missions. The stage restarted at T+27 min 38 sec for a 59 second long second burn that accelerated Bangabandhu 1 to GTO.

B1046 was test fired at McGregor, Texas during late February or early March 2018 and was briefly hot fired at LC 39A with the second stage but no payload attached, on May 4, 2018.

DM1 NASA Crew Dragon Premier

A Falcon 9 v1.2 Block 5 boosted the first SpaceX Crew Dragon into low Earth orbit on an uncrewed test flight from KSC on March 2, 2019. Liftoff from LC 39A took place at 07:49 UTC. The 13 foot diameter spacecraft, consisting of a 4.9 meter tall capsule and a 3.7 meter tall "trunk" sheathed in fixed solar arrays, was, at about 12.5 tonnes, the heaviest payload yet launched by a Falcon 9.

New first stage B1051.1 provided 771 metric tons of liftoff thrust. It shut down about 2.5 minutes after launch, then performed entry and landing burns to land downrange on Of Course I Still Love You (OCISLY). It was the 35th successful first stage landing. The second stage fired until reaching orbit about 9 minutes after liftoff.

Dragon DM-1 (Commercial Crew Program Demonstration Mission 1), carrying a dummy astronaut named "Ripley" and about 181 kg of cargo, docked with the International Space Station on March 3. It undocked March 8 before splashing down about 370 km east of Cape Canaveral.

B1051 was test fired at McGregor, Texas during August 2018. The entire vehicle with Dragon attached rolled out for mechanical fit checks at LC 39A on January 3, 2019. It rolled out again for a brief static test, again with Dragon attached, on January 24, 2019.



Starlink and Reuse

Crew Dragon DM2 NASA U.S. Crew Launch

NASA astronauts Doug Hurley and Bob Behnken flew to orbit in a SpaceX Crew Dragon atop a Falcon 9 v1.2 Block 5 from KSC 39A on May 30, 2020. It was the third spaceflight for both astronauts. The 19:22:45 UTC liftoff followed a weather scrub on May 27. The commercial Crew Dragon test flight to ISS was the first U.S.-launched crewed mission since Space Shuttle retired in 2011. Crew Dragon separated from the Falcon 9 second stage about 12 minutes after liftoff to begin its roughly 19-hour trip to dock with ISS.

First stage B1058.1 fired its nine Merlin 1D engines for 2 min 33 sec before shutting down and separating. The stage performed entry and landing burns before landing on the "Of Course I Still Love You" drone ship about 9 min 22 sec after liftoff. The second stage fired its single Merlin 1D Vacuum engine for 6 min 3 sec to reach a roughly 190 x 205 km x 51.6 deg orbit.

The first stage was static fired at McGregor, Texas, likely during August 2019. It performed a hot fire test at LC 39A on May 22, 2020, with Crew Dragon stacked atop the vehicle.

With the arrival of "Block 5", SpaceX began to launch its first Starlink missions to build the company's own Internet satellite constellation. The flights, which began on May 24, 2019, carried up to 60 satellites at a time, almost always using previously flown boosters. Starlink accounted for more than half of all Falcon 9 launches in 2020-22 and likely beyond. By mid-2022 two boosters, B1051 and B1060, had already launched 10 Starlink missions each.

Starlink drove launch rates and pushed booster reuse numbers. Only 29 of the 53 launched first stages were recovered during 2015-2018, but more than 96% of the next 107 had landed successfully by August 31, 2022. Five first stages had flown more than 10 times as of August 19, 2022. Two had flown 13 times. First stage recovery success meant that in 2021 only two new first stages were introduced during a year that saw 31 launches. About 10 to 12 first stages remained active at a time, typically cycling back to launch pads in one-to-three-month intervals. The company also refined its payload fairing reuse methods, allowing most launches to use previously flown fairing halves.

Vehicle Configurations

Payload
(metric tons)
r = downrange landing
z = RTLS to LZ

	185 km x (1) 28.5 deg (2) 98 deg (4) 51.6 deg (4r) 51.6 deg (4z) 51.6 deg	GTO 185 x35,788km x27 deg ~1,800 m/s from GEO	Escape Velocity (5) LEO +3,150 m/s (6) LEO +3,750 m/s	Height [1] Dragon [2] PLF	Liftoff Mass
Falcon 9 v1.0 Block 1 2010	9.0 t (1) 8.5 t (4)	3.4 t	2 t (5)	[1] 48.1 m	318 t
Falcon 9 v1.1 Block 2 2013	13.15 t (1)	4.85 t	~2.9 t (5)	[1] 63.3 m [2] 68.4 m	506 t
Falcon 9 v1.2 Block 3 2015	~17.4 t (1) 8.63 t (1r) 8.43 t (1z)	~6.4 t 5.28 t (r)	~3.8 (5) 0.57 (5r)	[1] 65.5 m [2] 69.98 m	556 t
Falcon 9 v1.2 Block 4 2017	~17.4 t (1) 8.6 t (1r) 8.4 t (4r)	6.76 t 3.72 t (r)		[1] 65.5 m [2] 69.98 m	556 t
Falcon 9 v1.2 Block 5 2018	22.8 t (1) 16.25t (4r)	8.3 t 5.3 t (r)	4.02 t (5) 0.61 t (5r)	[1] [2] 69.98 m	587 t

Vehicle Components

	Falcon 9 Stage 1 - v1.0 Merlin 1C	Falcon 9 Stage 2 v1.0 Merlin 1C	Falcon 9 Stage 1 v1.1 Merlin 1D	Falcon 9 Stage 2 v1.1 Merlin 1D	Falcon 9 Stage 1 v1.2 Block 3	Falcon 9 Stage 2 v1.2 Block 3
Dia. (m)	3.66 m	3.66 m	3.66 m	3.66 m	3.66 m	3.66 m
Len. (m)	~30.1 m no I/S	~10.0 m Incl I/S	~40.9 m no I/S	~14.6 m Incl I/S	~40.9 m no I/S	~16.0 m Incl I/S
Mp (t)	~239.3 t	~48.9 t	~385 t	~93 t	~411 t	~111.5 t
GLOW (t)	~258.5 t	~52 t	~404 t	~99 t	~438.2 t	~116 t
Merlin	1C	Vac	1D	1D Vac	1D FT	1D Vac FT
#Engs	9	1	9	1	9	1
Fuel	RP1	RP1	RP1	RP1	RP1	RP1
OX	LOX	LOX	LOX	LOX	LOX	LOX
T (SL)	385.825 t		600.109 t		694 t	
T (Vac)	442.938 t	42.18 t	680.396 t	81.647 t	757 t	95.255 t
ISP (SL)	266 s		282 s		283 s	
ISP (Vac)	304 s	336 s	311 s	340 s	312 s	348 s
tBurn (s)	180 s	346 s	185 s	375 s	162 s	397 s
Throttle						

Vehicle Components Continued

	Falcon 9 Stage 1 - v1.2 Block 5	Falcon 9 Stage 2 v1.2 Block 5	Falcon 9 PLF	Falcon 9	Falcon 9	Falcon 9
Dia. (m)	3.66 m	3.66 m	5.2 m			
Len. (m)	~40.9 m no I/S	~16.0 m Incl I/S	13.9 m			
Mp (t)	~418.7 t	~111.5 t				
GLOW (t)	~445.9 t	~116 t	~2.0 t			
Merlin	1D FT	1D Vac FT				
#Engs	9	1				
Fuel	RP1	RP1				
OX	LOX	LOX				
T (SL)	775.65 t					
T (Vac)	846 t	95.255 t				
ISP (SL)	283 s					
ISP (Vac)	312 s	348 s				
tBurn (s)	161.2 s	397 s				
Throttle						

Falcon 9 v1.1 and v1.2 Flight History

Date	Falcon 9 Variant/No.	Payload	Mass	Site	Orbit (kmxkmxdeg)
09/29/13	v1.1 F9-6	Cassiope/5 Cubesats	0.6	VA 4E	500x1500x80 LEO [8]
12/03/13	v1.1 F9-7	SES 8	3.183	CC 40	295x80000x20.8 GTO+[9]
01/06/14	v1.1 F9-8	Thaicom 6	3.016	CC 40	295x90000x22.5 GTO+[A]
04/18/14	v1.1 F9-9	CRS-3/Dragon 5	~7.76	CC 40	313x332x51.6 ISS [10]
07/14/14	v1.1 F9-10	Orbcomm OG2 (6sats)	1.032	CC 40	614x743x47 LEO [11]
08/05/14	v1.1 F9-11	Asiasat 8	4.535	CC 40	185x35786x24.3 GTO
09/07/14	v1.1 F9-13	Asiasat 6	4.428	CC 40	184x35762x25.3 GTO
09/21/14	v1.1 F9-12	CRS-4/Dragon 6	~7.716	CC 40	199x359x51.64 ISS
01/10/15	v1.1 F9-14	CRS-5/Dragon 7	~7.807	CC 40	206x353x51.6 ISS [12]
02/11/15	v1.1 F9-15	DSCOVER	0.57	CC 40	187x1371156x37 EEO [13]
03/02/15	v1.1 F9-16	Eutelsat 115WB/ABS 3A	4.159	CC 40	400x63300x24.8 GTO+
04/14/15	v1.1 F9-18	CRS-6/Dragon 8	~7.505	CC 40	199x364x51.65 ISS [14]
04/27/15	v1.1 F9-17	TurkmenAlem 52E	4.5	CC 40	180x36600x25.5 GTO
06/28/15	v1.1 F9-20	CRS-7/Dragon 9	~7.944	CC 40	[FTO] [15]
12/22/15	v1.2 F9-21	Orbcomm OG2	1.892	CC 40	620x660x47 LEO [16]
01/17/16	v1.1 F9-19	Jason 3	0.553	VA 4E	1305x1320x66 LEO [17]
03/04/16	v1.2 F9-22	SES 9	5.271	CC 40	290x40600x28 GTO [18]
04/08/16	v1.2 F9-23	CRS 8/Dragon 10	~8.626	CC 40	200x360x51.6 ISS [19]
05/06/16	v1.2 F9-24	JCSAT 14	4.696	CC 40	189x35957x23.7 GTO [20]
05/27/16	v1.2 F9-25	Thiacom 8	3.025	CC 40	350x90226x21.2 GTO+[21]
06/15/16	v1.2 F9-26	Eutelsat 117WB/ABS2A	~4.15	CC 40	395x62591x24.7 GTO+[22]
07/18/16	v1.2 F9-27	CRS 9/Dragon 11	~7.747	CC 40	200x360x51.6 ISS [23]
08/14/16	v1.2 F9-28	JCSAT 16	4.6	CC 40	184x35912x20.9 GTO [21]
09/01/16	v1.2 F9-29	AMOS 6	5.5	CC 40	[PAD] [24]
01/14/17	v1.2 F9-30	Iridium NEXT 1	8.6	VA 4E	667x86.4 LEO [25]
02/19/17	v1.2 F9-32	CRS-10/Dragon 12	~8.43	KC 39A	209x363x51.6 ISS [26]
03/16/17	v1.2 F9-31	EchoStar 23	5.6	KC 39A	179x35903x22.4 GTO [27]
03/30/17	v1.2 F9-33	SES 10	5.282	KC 39A	217x33395x26.3 GTO [28]
05/01/17	v1.2 F9-34	NROL 76	~2.8?	KC 39A	400x51? LEO? [29]
05/15/17	v1.2 F9-35	Inmarsat 5 F4	6.086	KC 39A	381x69839x24.5 GTO+[30]
06/03/17	v1.2 F9-36	CRS-11/Dragon 6.2	~8.198	KC 39A	~210x360x51.6 ISS [31]
06/23/17	v1.2 F9-37	BulgariaSat 1	3.669	KC 39A	210x65640x23.9 GTO+[32]
06/25/17	v1.2 F9-38	Iridium Next 2	8.60	VA 4E	625x86.4 LEO [33]
07/05/17	v1.2 F9-39	Intelsat 35e	6.761	KC 39A	296x42742x25.6 GTO [34]
08/14/17	v1.2 F9-41	CRS-12/Dragon 13	~8.4	KC 39A	210x360x51.6 ISS [35]
08/24/17	v1.2 F9-40	Formosat 5	0.475	VA 4E	720x98.28 SSO [36]
09/07/17	v1.2 F9-42	OTV-5 (X-37B)	~5.00	KC 39A	350x43 LEO [37]
10/09/17	v1.2 F9-44	Iridium NEXT 3	8.6	VA 4E	625x86.6 LEO [38]
10/11/17	v1.2 F9-43	EchoStar 105/SES 11	5.2	KC 39A	309x40519x27.9 GTO [39]
10/30/17	v1.2 F9-45	Koreasat 5A	3.7	KC 39A	285x50185x22.0 GTO [40]
12/15/17	v1.2 F9-47	CRS-13/Dragon 8.2	~7.7	CC 40	210x360x51.6 ISS [41]
12/23/17	v1.2 F9-48	Iridium NEXT 4	8.6	VA 4E	625x86.6 deg LEO [42]
01/08/18	v1.2 F9-46	Zuma		CC 40	1000x50? LEO? [43]
01/31/18	v1.2 F9-49	Govsat 1	4.230	CC 40	250x52000x27 GTO [44]
02/22/18	v1.2 F9-50	Paz/Microsat 2a/b	~2.000	VA 4E	514x97.4 SSO [45]
03/06/18	v1.2 F9-51	Hispasat 30W-6	6.092	CC 40	184x22261x26.9 GTO-[46]
03/30/18	v1.2 F9-52	Iridium NEXT 5	8.6	VA 4E	625x86.6 deg LEO [47]
04/02/18	v1.2 F9-53	Dragon 10.2/CRS-14	~8.125	CC 40	210x360x51.6 ISS [48]
04/18/18	v1.2 F9-54	TESS	0.35	CC 40	299x299450x29.5 EEO [49]
05/11/18	v1.2 F9-55	Bangabandhu 1	~3.72	KC 39A	308x35549x19.3 GTO [50]
05/22/18	v1.2 F9-56	Iridium 6/GRACE-FO	~5.5	VA 4E	494x710x86.7 LEO [51]
06/04/18	v1.2 F9-57	SES 12	5.384	CC 40	250x58370x26 GTO+[52]

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06/29/18	v1.2	F9-58	Dragon 11.2/CRS 15	~8.175	CC 40	227x387x51.64	ISS [53]
07/22/18	v1.2	F9-59	Telstar 19V	7.075	CC 40	243x17863x27	GTO-[54]
07/25/18	v1.2	F9-60	Iridium NEXT 7	8.6	VA 4E	600x625x86.7	LEO [55]
08/07/18	v1.2	F9-61	Merah Putih	5.8	CC 40	193x29503x27	GTO-[56]
09/10/18	v1.2	F9-62	Telstar 18V	7.06	CC 40	259x18060x26.95	GTO-[57]
10/08/18	v1.2	F9-63	SAOCOM 1A	3.0	VA 4E	616x635x97.9	SSO [58]
11/15/18	v1.2	F9-64	Es'hail 2	~5.3	KC 39A	201x37866x25	GTO [59]
12/03/18	v1.2	F9-65	Spaceflight SSO-	~4.0	VA 4E	573x589x97.8	SSO [60]
12/05/18	v1.2	F9-66	CRS-16/Dragon 12.2	~8.039	CC 40	204x358x51.6	ISS [61]
12/23/18	v1.2	F9-67	GPS 3-1	4.4	CC 40	1200x20200x55	MTO [62]
01/11/19	v1.2	F9-68	Iridium NEXT 8	8.6	VA 4E	600x625x86.7	LEO [63]
02/22/19	v1.2	F9-70	Nusantara Satu+usats	4.85	CC 40	260x69036x27.55	GTO+[64]
03/02/19	v1.2	F9-69	DM-1/Crew Dragon	~12.5	KC 39A	194x358x51.6	ISS [65]
05/04/19	v1.2	F9-71	CRS-17/Dragon 13.2	~8.0	CC 40	200x360x51.6	ISS [66]
05/24/19	v1.2	F9-72	Starlink 0.9 (60 ea)	13.608	CC 40	440x440x53	LEO [67]
06/12/19	v1.2	F9-73	RCM	4.29	VA 4E	600x600x97.74	SSO [68]
07/25/19	v1.2	F9-74	CRS-18/Dragon 8.3	~7.831	CC 40	200x360x51.6	ISS [69]
08/06/19	v1.2	F9-75	AMOS 17	6.5	CC 40	221x35750x26.1	GTO [70]
11/11/19	v1.2	F9-76	Starlink 1 (60 sat)	15.60	CC 40	280x280x53	LEO [71]
12/05/19	v1.2	F9-77	CRS-19/Dragon 6.3	~7.804	CC 40	200x360x51.6	ISS [72]
12/17/19	v1.2	F9-78	JCSat 18/Kacific 1	6.956	CC 40	273x20324x26.9	GTO-[73]
01/07/20	v1.2	F9-79	Starlink 1 F2	15.6	CC 40	290x290x53	LEO [74]
01/19/20	v1.2	F9-80	IFA/Crew Dragon 2	~12.5	KC 39A		<SUB> [75]
01/29/20	v1.2	F9-81	Starlink 1 F3	15.6	CC 40	290x290x53	LEO [76]
02/17/20	v1.2	F9-82	Starlink 1 F4	15.6	CC 40	216x386x53	LEO [77]
03/07/20	v1.2	F9-83	CRS-20/Dragon 12.3	~7.259	CC 40	204x384x51.6	ISS [78]
03/18/20	v1.2	F9-84	Starlink 1 F5	15.6	KC 39A	213x350x53	LEO [79]
04/22/20	v1.2	F9-85	Starlink 1 F6	15.6	KC 39A	213x367x53	LEO [80]
05/30/20	v1.2	F9-87	DM-2/Crew Dragon	~12.5	KC 39A	190x205x51.6	ISS [81]
06/04/20	v1.2	F9-86	Starlink 1 F7	15.6	CC 40	190x380x53	LEO [82]
06/13/20	v1.2	F9-88	Starlink 1 F8	15.41	CC 40	190x380x53	LEO [83]
06/30/20	v1.2	F9-90	GPS 3-3	4.311	CC 40	400x20200x55	MTO [84]
07/20/20	v1.2	F9-91	Anasis 2		CC 40	211x45454x27.4	GTO+[85]
08/07/20	v1.2	F9-89	Starlink 1 F9	15.192	KC 39A	380x400x53	LEO [86]
08/18/20	v1.2	F9-92	Starlink 1-10/Skysat	15.44	CC 40	220x380x53	LEO [87]
08/30/20	v1.2	F9-93	SAOCOM 1B	3.05	CC 40	610x610x97.9	SSO [88]
09/03/20	v1.2	F9-94	Starlink 1-11	15.6	KC 39A	220x380x53	LEO [89]
10/06/20	v1.2	F9-95	Starlink 1-12	15.6	KC 39A	260x280x53	LEO [90]
10/18/20	v1.2	F9-97	Starlink 1-13	15.6	KC 39A	260x280x53.1	LEO [91]
10/24/20	v1.2	F9-98	Starlink 1-14	15.6	CC 40	260x260x53	LEO [92]
11/05/20	v1.2	F9-96	GPS 3-4	4.311	CC 40	420x22440x55	MTO [93]
11/16/20	v1.2	F9-99	Crew 1/Crew Dragon	~12.5	KC 39A	190x210x51.6	ISS [94]
11/21/20	v1.2	F9-100	Sentinel 6	1.192	VA 4E	1336x1336x66	LEO [95]
11/25/20	v1.2	F9-101	Starlink 1-15	15.6	CC 40	210x360x53	LEO [96]
12/06/20	v1.2	F9-102	CRS-21/Cargo Dragon	~12.5	KC 39A	192x211x51.7	ISS [97]
12/13/20	v1.2	F9-103	SXM-7	~7.0	CC 40	249x19515x27	GTO-[98]
12/19/20	v1.2	F9-104	NROL-108		KC 39A	400x50?	LEO? [99]
01/08/21	v1.2	F9-105	Turksat 5A	3.5	CC 40	286x55031x17.66	GTO+[100]
01/20/21	v1.2	F9-106	Starlink 1-16	15.6	KC 39A	260x280x53.1	LEO [101]
01/24/21	v1.2	F9-107	Transporter 1	~3.57	CC 40	532x538x98.5	SSO [102]
02/04/21	v1.2	F9-109	Starlink 1-18	15.6	CC 40	260x280x53.1	LEO [103]
02/16/21	v1.2	F9-110	Starlink 1-19	15.6	CC 40	250x280x53	LEO [104]
03/04/21	v1.2	F9-108	Starlink 1-17	15.6	KC 39A	250x290x53	LEO [105]
03/11/21	v1.2	F9-111	Starlink 1-20	15.6	CC 40	250x280x53	LEO [106]
03/14/21	v1.2	F9-112	Starlink 1-21	15.6	KC 39A	250x290x53	LEO [107]
03/24/21	v1.2	F9-113	Starlink 1-22	15.6	CC 40	250x280x53	LEO [108]
04/07/21	v1.2	F9-114	Starlink 1-23	15.6	CC 40	270x53	LEO [109]

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04/23/21	v1.2	F9-115	Crew 2/Crew Dragon	~12.5	KC 39A	190x210x51.6	ISS [110]
04/29/21	v1.2	F9-116	Starlink 1-24	15.6	CC 40	250x280x53	LEO [111]
05/04/21	v1.2	F9-117	Starlink 1-25	15.6	KC 39A	259x284x53	LEO [112]
05/09/21	v1.2	F9-118	Starlink 1-27	15.6	CC 40	259x284x53	LEO [113]
05/15/21	v1.2	F9-119	Starlink 1-26	~13.7	KC 39A	581x569x53	LEO [114]
05/26/21	v1.2	F9-120	Starlink 1-28	15.6	CC 40	260x300x53	LEO [115]
06/03/21	v1.2	F9-121	CRS-22/Cargo Dragon	~12.5	KC 39A		ISS [116]
06/06/21	v1.2	F9-122	SXM-8	~7.0	CC 40	235x19379x26.49	GTO-[117]
06/17/21	v1.2	F9-123	GPS 3-5	4.311	CC 40	394x20176x55.0	MTO [118]
06/30/21	v1.2	F9-124	Transporter 2	~2.4	CC 40	515x536x97.5	SSO [119]
08/29/21	v1.2	F9-125	CRS-23/Cargo Dragon	~12.5	KC 39A		ISS [120]
09/14/21	v1.2	F9-126	Starlink 2-1	~13.26	VA 4E	215 x 70	LEO [121]
09/16/21	v1.2	F9-127	Inspiration 4	~12.5	KC 39A	190x575x51.6	LEO [122]
11/11/21	v1.2	F9-128	Crew 3/Crew Dragon	~12.5	KC 39A	190x210x51.6	ISS [123]
11/13/21	v1.2	F9-129	Starlink 4-1	~15.5?	CC 40	210x340x53.2	LEO [124]
11/24/21	v1.2	F9-130	DART	0.61	VA 4E	0.94x1.07AUx3.8	HCO [125]
12/02/21	v1.2	F9-131	Starlink 4-3	~14.0?	CC 40	430x53.2	LEO [126]
12/09/21	v1.2	F9-132	IXPE	0.337	KC 39A	588x603x0.2	LEO [127]
12/18/21	v1.2	F9-133	Starlink 4-4	~14.0?	VA 4E	210x340x53.22	LEO [128]
12/19/21	v1.2	F9-134	Turksat 5B	4.5	CC 40	198x68931x27.12	GTO+[129]
12/21/21	v1.2	F9-135	CRS-24/Cargo Dragon	~13.5	KC 39A	195x200x51.6	ISS [130]
01/06/22	v1.2	F9-136	Starlink 4-5	~14.0	KC 39A	210x339x53.22	LEO [131]
01/13/22	v1.2	F9-137	Transporter 3	~2.5	CC 40	535x97.5	SSO [132]
01/19/22	v1.2	F9-138	Starlink 4-6	~14.0	KC 39A	210x339x53.22	LEO [133]
01/31/22	v1.2	F9-139	CSG FM2	2.2	CC 40	620x97.86	SSO [134]
02/02/22	v1.2	F9-140	NROL-87		VA 4E	512x97.4?	SSO [135]
02/03/22	v1.2	F9-141	Starlink 4-7	~14.0	KC 39A	210x339x53.2	LEO [136]
02/21/22	v1.2	F9-142	Starlink 4-8	~13.1	CC 40	325x337x53.2	LEO [137]
02/25/22	v1.2	F9-143	Starlink 4-11	~14.2	VA 4E	306x316x53.2	LEO [138]
03/03/22	v1.2	F9-144	Starlink 4-9	~13.4	KC 39A	305x317x53.2	LEO [139]
03/09/22	v1.2	F9-145	Starlink 4-10	~13.68	CC 40	305x317x53.2	LEO [140]
03/19/22	v1.2	F9-146	Starlink 4-12	16.25	CC 40	304x318x53.2	LEO [141]
04/02/22	v1.2	F9-147	Transporter 4	~2.0	CC 40	650&500x97.4	SSO [142]
04/08/22	v1.2	F9-148	Axiom 1/Crew Dragon	~12.5	KC 39A	197x210x51.6	ISS [143]
04/17/22	v1.2	F9-149	NROL 85 (2xIntruder)	~6.5	VA 4E	1024x1220x64.5	LEO [144]
04/21/22	v1.2	F9-150	Starlink 4-14	~15.9	CC 40	304x318x53.2	LEO [145]
04/27/22	v1.2	F9-151	Crew 4/Crew Dragon	~12.5	KC 39A	195x210x51.6	ISS [146]
04/29/22	v1.2	F9-152	Starlink 4-16	~15.9	CC 40	304x317x53.2	LEO [147]
05/06/22	v1.2	F9-153	Starlink 4-17	~15.9	KC 39A	305x317x53.2	LEO [B]
05/13/22	v1.2	F9-154	Starlink 4-13	~15.9	VA 4E	306x316x53.2	LEO [148]
05/14/22	v1.2	F9-155	Starlink 4-15	~15.9	CC 40	305x318x53.2	LEO [149]
05/18/22	v1.2	F9-156	Starlink 4-18	~15.9	KC 39A	304x318x53.2	LEO [150]
05/25/22	v1.2	F9-157	Transporter 5	~2.0	CC 40	414x532x97.51	SSO [151]
06/08/22	v1.2	F9-158	Nilesat 301	3.938	CC 40	310x44880x18.6	GTO+[152]
06/17/22	v1.2	F9-159	Starlink 4-19	~15.9	KC 39A	232x337x53.2	LEO [153]
06/18/22	v1.2	F9-160	SARah 1	~4.0	VA 4E	750x98.4	SSO [154]
06/19/22	v1.2	F9-161	Globalstar FM15	~0.7	CC 40	1200x52	LEO [155]
06/29/22	v1.2	F9-162	SES 22	3.5	CC 40	310x35627x16.54	GTO [156]
07/07/22	v1.2	F9-163	Starlink 4-21	~15.9	CC 40		LEO [157]
07/11/22	v1.2	F9-164	Starlink 3-1	~13.8	VA 4E	310x320x97.6	SSO [158]
07/15/22	v1.2	F9-165	CRS-25/Cargo Dragon	~13.5	KC 39A	195x200x51.6	ISS [159]
07/17/22	v1.2	F9-166	Starlink 4-22	~15.9	CC 40	232x338x53.2	LEO [160]
07/22/22	v1.2	F9-167	Starlink 3-2	~13.8	VA 4E	308x321x97.6	SSO [161]
07/24/22	v1.2	F9-168	Starlink 4-25	~15.9	KC 39A	232x338x53.2	LEO [162]
08/04/22	v1.2	F9-169	KPLO	0.678	CC 40	245x1239000x30	EEO [163]
08/10/22	v1.2	F9-170	Starlink 4-26	~15.6	KC 39A	230x335x53.2	LEO [164]
08/12/22	v1.2	F9-171	Starlink 3-3	~12.9	VA 4E	308x321x97.6	SSO [165]

08/19/22	v1.2	F9-172	Starlink 4-27	~15.9	CC 40	232x336x53.2	LEO [166]
08/28/22	v1.2	F9-173	Starlink 4-23	16.7	CC 40	232s336x53.2	LEO [167]
08/31/22	v1.2	F9-174	Starlink 3-4	~13.8	VA 4E	307x321x97.6	SSO [168]

- [8] First Falcon 9 v1.1. First VAFB SLC 4E launch of Falcon 9.
1st stage performed two reentry burns (3 and 1 engine), but 2nd burn cutoff early due high roll rates. 2nd stage restart for disposal burn failed.
- [9] First Falcon 9 GTO+ launch. Targeted 295 x 80,000 km x 20.75 degree supersynchronous transfer orbit. Stg 1 briefly restarted post sep. Fire reported in Stg1 octaweb during ascent.
- [A] Lower than planned fuel reserves reported at end of final Stg2 burn.
Planned orbit achieved.
- [10] First Falcon 9 fitted with extending landing legs. First stage performed two retro burns after separation, lowering itself to a simulated landing in the Atlantic off the Georgia/S. Carolina coast.
- [11] 2nd Falcon 9 with legs. First stage performed two retro burns and landed in Atlantic but exploded during tip over.
- [12] 1st stg attempted landing on converted barge about 320 km downrange, but landed hard on barge and was lost.
- [13] 187 x 1,371,156 km x 37 degree insertion orbit. DSCVR bound for Earth-Sun L1. Stg 1 barge landing attempt abandoned due high seas.
- [14] First stage landed hard on downrange landing platform and was destroyed.
- [15] Broke up at about T+2m 19sec, before staging, due Stg2 LOX tank overpress.
- [16] First stage boosted back to CC LZ-1 (former LC 13) and landed. Performed boostback, reentry, and landing burns using 3, 1, and 1 engine. First Falcon 9 v1.2 (Full Thrust) flight.
- [17] First stage landed on downrange landing platform, but one leg failed to lock in place. Stage fell over and was destroyed.
- [18] First stage landing on downrange platform failed.
- [19] 3,136 kg cargo, incl 1,413 kg BEAM in trunk. 1st stg landed on barge (1st barge success).
- [20] First stage landed on downrange platform. First GTO landing. First successful 3-engine landing.
- [21] First stage landed on downrange platform.
- [22] First stage destroyed during landing attempt on downrange platform. One of three engines produced low thrust during final landing burn. Stage "accordianed" on hard landing. Mission otherwise successful.
- [23] First stage landed at CC LZ-1.
- [24] F9 and AMOS 6 destroyed in explosion during hot fire countdown at SLC 40.
Launch was planned for 09/03/16.
- [25] First stage (B1029) landed downrange on drone ship "Just Read the Instructions".
- [26] First stage (B1031) landed at CC LZ-1.
- [27] First stage (B1030) purposely expended. No legs or fins. First expendable v1.2. Allowed heaviest-yet GTO payload.
- [28] First Stg 1 reflight using B1021. Stage landed downrange on drone ship "Of Course I Still Love You". PLF half recovery test.
Planned 218 x 35,410 km x 26.2 deg, but achieved agreed parameters.
- [29] First stage (B1032) landed at CC LZ-1. Block 4 second stage.
- [30] Expendable first stage (B1034). Heaviest-yet GTO payload. Stg 2 burned to depletion. Block 4 second stage.
- [31] First stage (B1035) landed at CC LZ-1. 1st reflight by Dragon C106.
- [32] First stage (B1029.2) landed OCISLY. 2nd Stg 1 reflight.
- [33] First stage (B1036) landed JRTI.

- [34] Expendable first stage (B1037). Heaviest-yet GTO payload. Stg 2 burned to depletion. Block 4 second stage.
- [35] First stage B1039 landed at CC LZ-1. First Blk 4 Stg 1.
- [36] First stage B1038 landed JRTI.
- [37] First stage B1040 landed CC LZ-1. Blk 4 Stg 1&2.
- [38] First stage B1041 landed JRTI.
- [39] First stage B1031.2 landed on OCISLY downrange.
- [40] First stage B1042 landed on OCISLY downrange.
- [41] First stage B1035.2 landed LZ-1.
- [42] First stage B1036.2 expended.
- [43] First stage B1043 landed LZ-1.
- [44] First stage B1032.2 expended. Insertion orbit estimated. Actual orbit not published.
- [45] First stage B1038.2 expended. 1st PLF 2.0, recovery attempted.
- [46] First stage B1044.1 expended. High seas prevented recovery.
- [47] First stage B1041.2 expended. 2nd PLF 2.0 recovery attempt (failed).
- [48] First stage B1039.2 expended.
- [49] First stage B1045 (last new Block 4) landed on OCISLY downrange.
- [50] First stage B1046 (first Block 5, with Block 5 Stg 2) landed OSCILY.
- [51] First stage B1043.2 expended. PLF recovery failed. GRACE-FO to 483 x 505 km x 86.4 deg, then S2 2nd burn to final orbit.
- [52] First stage B1040.2 expended.
- [53] First stage B1045.2 expended.
- [54] 2nd Blk 5. B1047 landed OCISLY.
- [55] First stage B1048 landed JRTI
- [56] First stage B1046.2 landed OCISLY.
- [57] First stage B1049 landed OCISLY.
- [58] First stage B1048.2 landed LZ-4 (1st LZ 4)
- [59] B1047.2 landed OCISLY.
- [60] B1046.3 landed JRTI. 1st 3rd reflight.
- [61] B1050.1 rolled after entry burn due grid fin hyd. failure. Stage fell short of LZ-1, impacted Atlantic ~2miles offshore.
- [62] B1054.1 expended.
- [63] First stage B1049.2 landed JRTI
- [64] First stage B1048.3 landed OCISLY
- [65] First stage B1051.1 landed OCISLY
- [66] First stage B1056.1 landed OCISLY
- [67] First stage B1049.3 landed OCISLY
- [68] First stage B1051.2 landed LZ-4
- [69] First stage B1056.2 landed LZ-1
- [70] First stage B1047.3 expended
- [71] First stage B1048.4 landed OCISLY
- [72] First stage B1059.1 landed OCISLY (S2 long coast test)
- [73] First stage B1056.3 landed OCISLY
- [74] First stage B1049.4 landed OCISLY
- [75] Crew Dragon In-Flight Abort Test. S1 B1046.4 expended.
- [76] First stage B1051.3 landed OCISLY
- [77] First stage B1056.4 missed OCISLY
- [78] First stage B1059.2 landed LZ-1
- [79] First stage B1048.5 lost 1 engine during ascent, no OCISLY landing
- [80] First stage B1051.4 landed OCISLY
- [81] First Crew Launch. First stage B1058.1 landed OCISLY
- [82] First stage B1049.5 landed JRTI. First "fifth" landing.
- [83] First stage B1059.3 landed OCISLY. Skysats 16-18 also orbited.
- [84] First stage B1060.1 landed JRTI.
- [85] First stage B1058.2 landed JRTI.
- [86] First stage B1051.5 landed OCISLY. BlackSky Global 7/8 also orbited.

- [87] First stage B1049.6 landed OCISLY. Skysats 19-21 also orbited.
- [88] First stage B1059.4 landed LZ-1. 1st CC LEO/S since 1969.
- [89] First stage B1060.2 landed OCISLY
- [90] First stage B1058.3 landed OCISLY
- [91] First stage B1051.6 landed OCISLY
- [92] First stage B1060.3 landed JRTI
- [93] First stage B1062.1 landed OCISLY
- [94] First stage B1061.1 landed JRTI
- [95] First stage B1063.1 landed LZ-4
- [96] First stage B1049.7 landed OCISLY
- [97] First stage B1058.4 landed OCISLY. First Dragon 2 Cargo. C208-1.
- [98] First stage B1051.7 landed JRTI
- [99] First stage B1059.5 landed LZ-1
- [100] First stage B1060.4 landed JRTI
- [101] First stage B1051.8 landed JRTI
- [102] First stage B1058.5 landed OCISLY, 2nd F9 LEO/S from CC
- [103] First stage B1060.5 landed OCISLY
- [104] First stage B1059.6 failed to land OCISLY. One engine shut down during ascent in final seconds due boot failure and overheating. Overheating affected entry burn, which ended with the engine compartment on fire.
- [105] First stage B1049.8 landed OCISLY.
- [106] First stage B1058.6 landed JRTI.
- [107] First stage B1051.9 landed OCISLY.
- [109] First stage B1058.7 landed OCISLY.
- [110] First stage B1061.2 landed OCISLY.
- [111] First stage B1060.7 landed JRTI.
- [112] First stage B1049.9 landed OCISLY.
- [113] First stage B1050.10 landed JRTI.
- [114] First stage B1058.8 landed OCISLY. 52 Starlink + Capella 6 + Tyvak-0130 Rideshares.
- [115] First stage B1063.2 landed JRTI.
- [116] First stage B1067.1 landed OCISLY.
- [117] First stage B1061.3 landed JRTI.
- [118] First stage B1062.2 landed JRTI.
- [119] First stage B1060.8 landed LZ-1.
- [120] First stage B1061.4 landed ASOG. C208-2.
- [121] First stage B1049.10 landed OCISLY.
- [122] First stage B1062.3 landed JRTI. 1st all-private orbital flight. Crew Dragon C207.2.
- [123] Stg 1 B1067.2 landed ASOG. Crew Dragon C210.1.
- [124] Stg 1 B1058.9 landed JRTI.
- [125] Stg 1 B1063.3 landed OCISLY.
- [126] Stg 1 B1060.9 landed ASOG. 48 Starlinks, 2 BlackSky Global.
- [127] Stg 1 B1061.5 landed JRTI. First F9 LEO/Equatorial mission.
- [128] Stg 1 B1051.11 landed OCISLY.
- [129] Stg 1 B1067.3 landed ASOG.
- [130] Stg 1 B1069.1 landed JRTI. Stg damaged during recovery.
- [131] Stg 1 B1062.4 landed ASOG.
- [132] Stg 1 B1058.10 landed LZ-1.
- [133] Stg 1 B1060.10 landed ASOG.
- [134] Stg 1 B1052.3 landed LZ-1.
- [135] Stg 1 B1071.1 landed LZ-4
- [136] Stg 1 B1061.6 landed ASOG. Solar storm caused at least 37 of 49 satellites to fall from orbit.
- [137] Stg 1 B1058.11 landed ASOG.
- [138] Stg 1 B1063.4 landed OCISLY.

[139] Stg 1 B1060.11 landed JRTI.
 [140] Stg 1 B1052.4 landed ASOG.
 [141] Stg 1 B1051.12 landed JRTI. 53 sats. 16.25t = heaviest payload
 [142] Stg 1 B1061.7 landed JRTI.
 [143] Stg 1 B1062.5 landed ASOG. 4 Crew on C206.3 "Endeavour". First
 Axiom private crew mission to ISS.
 [144] Stg 1 B1071.2 landed LZ-4.
 [145] Stg 1 B1060.12 landed JRTI.
 [146] Stg 1 B1067.4 landed ASOG. Crew-4 on C212.1 "Freedom".
 [147] Stg 1 B1062.6 landed JRTI.
 [B] Stg 1 B1058.12 landed ASOG.
 [148] Stg 1 B1063.5 landed OCISLY. 53 sats.
 [149] Stg 1 B1073.1 landed JRTI.
 [150] Stg 1 B1052.5 landed ASOG. 53 sats. Typ 2-burn ascent.
 [151] Stg 1 B1061.8 landed LZ-1. 59 sats. 2-burn ascent.
 [152] Stg 1 B1062.7 landed JRTI.
 [153] Stg 1 B1060.13 landed ASOG. 53 sats. 1-burn ascent. 50th F9
 or FH launch from LC 39A. 100th successful stage reuse.
 [154] Stg 1 B1071.3 landed LZ-4. SAR for German defense.
 [155] Stg 1 B1061.9 landed JRTI. Secret sat to 533 km x 54 deg, then
 two more Stg 2 burns to FM15 1200 km x 52 deg orbit.
 [156] Stg 1 B1073.2 landed ASOG.
 [157] Stg 1 B1058.13 landed JRTI. 53 sats. 1-burn ascent.
 [158] Stg 1 B1063.6 landed OCISLY. 46 sats. 2-burn ascent.
 [159] 00:44 UTC. Dragon 2 C208-3. B1067.5 landed ASOG. Boost-back.
 [160] B1051.13 landed JRTI. 53 sats. 1-burn ascent.
 [161] 17:39 UTC. B1071.4 landed OCISLY. 46 sats. 2-burn ascent.
 T-46s Abort previous day due engine back-up valve position.
 [162] B1062.8 landed ASOG. 53 sats. 1-burn ascent. Static 7/22/22.
 [163] 23:08 UTC. Korean Pathfinder Lunar Orbiter to
 245 x 1,239,000 km x 29.6 deg. Fifth orbital launch of 8/4/22,
 a Space-Age record. Second orbital launch from Cape Canaveral
 on 8/4/22. B1052.6 landed JRTI. To ballistic lunar transfer
 orbit.
 [164] B1073.3 landed ASOG. 52 sats. 1-burn ascent.
 [165] B1061.10 landed OCISLY. 46 sats. 2-burn ascent.
 [166] B1062.9 landed ASOG. 53 sats. 1-burn ascent.
 [167] 03:41 UTC. B1069.2, rebuilt with 9 new engines, landed ASOG.
 Heaviest-ever Falcon 9 payload. Improved Merlin Vacuum engine.
 [168] 05:50 UTC. B1063.7 landed OCISLY. 46 sats. 2-burn ascent.

LIST BY STAGE 1 SERIAL NUMBER

X = Expended
 OL = Ocean Landing
 DRL = Down Range Platform Landing
 LZ1 = Landing Zone 1 Landing
 -X = Failed Landing
 -S = Successful Landing (Scrapped)
 -D = Successful Landing (Saved for Display)
 -M = Successful Landing (Mothballed)
 STA = Structural Test Article
 QTA = Qualification Test Article

Stage 1 No.	Date	Falcon 9 Variant/No.	Description	Mass	Site	Stgl1/Result	Orbit
B0001	2007	v1.0	STA				
B0002	2007	v1.0	QTA/Grasshopper				
B0003	06/04/10	v1.0 F9-1	Dragon Qual Unit	~5.5	CC 40	X	LEO
B0004	12/08/10	v1.0 F9-2	Dragon C1	~5.5	CC 40	X	LEO
B0005	05/22/12	v1.0 F9-3	Dragon C2+	~6.02	CC 40	X	ISS
B0006	10/08/12	v1.0 F9-4	Dragon 3/CRS-1	~6.4	CC 40	X	[ISS]
B0007	03/01/13	v1.0 F9-5	Dragon 4/CRS-2	~6.54	CC 40	X	ISS
B1001	2013	v1.1	STA				
B1002	2013	v1.1	QTA/F9R Dev1			X	
B1003	09/29/13	v1.1 F9-6	Cassiope/5 Cubesats	0.6	VA 4E	OL-X	LEO
B10??	2013	v1.1	F9R Dev2 (not flown)				
B10??	12/03/13	v1.1 F9-7	SES 8	3.183	CC 40	X	GTO+
B100?	01/06/14	v1.1 F9-8	Thaicom 6	3.016	CC 40	X	GTO+
B10??	04/18/14	v1.1 F9-9	Dragon 5/CRS-3	~7.76	CC 40	OL-X	ISS
B10??	07/14/14	v1.1 F9-10	Orbcomm OG2 (6sats)	1.032	CC 40	OL-X	LEO
B10??	08/05/14	v1.1 F9-11	Asiasat 8	4.535	CC 40	X	GTO
B1010	09/21/14	v1.1 F9-12	Dragon 6/CRS-4	~7.716	CC 40	OL-X	ISS
B1011	09/07/14	v1.1 F9-13	Asiasat 6	4.428	CC 40	X	GTO
B1012	01/10/15	v1.1 F9-14	Dragon 7/CRS-5	~7.807	CC 40	DRL-X	ISS
B1013	02/11/15	v1.1 F9-15	DSCOVR	0.57	CC 40	OL-X	EEO
B1014	03/02/15	v1.1 F9-16	Eutelsat 115WB/ABS 3A	4.159	CC 40	X	GTO+
B1015	04/14/15	v1.1 F9-18	Dragon 8/CRS-6	~7.505	CC 40	DRL-X	ISS
B1016	04/27/15	v1.1 F9-17	TurkmenAlem 52E	4.5	CC 40	X	GTO
B1017	01/17/16	v1.1 F9-19	Jason 3	0.553	VA 4E	DRL-X	LEO
B1018	06/28/15	v1.1 F9-20	Dragon 9/CRS-7	~7.944	CC 40	X	[FTO]
B1019	12/22/15	v1.2 F9-21	Orbcomm OG2	1.892	CC 40	LZ1-D	LEO
B1020	03/04/16	v1.2 F9-22	SES 9	5.271	CC 40	DRL-X	GTO
B1021	04/08/16	v1.2 F9-23	Dragon 10/CRS 8	~8.626	CC 40	DRL	ISS
B1021.2	03/30/17	v1.2 F9-33	SES 10	5.282	KC 39A	DRL-D	GTO
B1022	05/06/16	v1.2 F9-24	JCSAT 14	4.696	CC 40	DRL	GTO
B1022.2	2016	v1.2	TX Hot Fire Tests				
B1023	05/27/16	v1.2 F9-25	Thiacom 8	3.025	CC 40	DRL	GTO+
B1023.2	02/06/18	FH FH-1	FH Demo Side		KC 39A	LZ1	HCO
B1024	06/15/16	v1.2 F9-26	Eutelsat 117WB/ABS2A	~4.15	CC 40	DRL-X	GTO+
B1025	07/18/16	v1.2 F9-27	Dragon 11/CRS 9	~7.747	CC 40	LZ1	ISS
B1025.2	02/06/18	FH FH-1	FH Demo Side		KC 39A	LZ1	HCO
B1026	08/14/16	v1.2 F9-28	JCSAT 16	4.6	CC 40	DRL-S	GTO
B1027	2016	FH FH-1	FH Core STA				
B1028	09/01/16	v1.2 F9-29	AMOS 6	5.5	CC 40	X	[PAD]
B1029	01/14/17	v1.2 F9-30	Iridium Next 1-10	8.6	VA 4E	DRL	LEO
B1029.2	06/23/17	v1.2 F9-37	BulgariaSat 1	3.669	KC 39A	DRL-D	GTO+
B1030	03/16/17	v1.2 F9-31	EchoStar 23	5.6	KC 39A	X	GTO

B1031	02/19/17	v1.2	F9-32	Dragon 12/CRS-10	~8.43	KC 39A	LZ1	ISS
B1031.2	10/11/17	v1.2	F9-43	EchoStar 105/SES 11	5.2	KC 39A	DRL	GTO
B1032	05/01/17	v1.2	F9-34	NROL 76	~2.8?	KC 39A	LZ1	LEO
B1032.2	01/31/18	v1.2	F9-49	GovSat 1	4.23	CC 40	X	GTO
B1033	02/06/18	FH	FH-1	FH Demo Core		KC 39A	DRL-X	HCO
B1034	05/15/17	v1.2	F9-35	Inmarsat 5 F4	6.086	KC 39A	X	GTO+
B1035	06/03/17	v1.2	F9-36	Dragon 6.2/CRS-11	~8.198	KC 39A	LZ1	ISS
B1035.2	12/15/17	v1.2	F9-47	Dragon 8.2/CRS-13	~7.7	CC 40	LZ1	ISS
B1036	06/25/17	v1.2	F9-38	Iridium Next 11-20	8.60	VA 4E	DRL	LEO
B1036.2	12/23/17	v1.2	F9-48	Iridium Next 4	8.6	VA 4E	X	LEO
B1037	07/05/17	v1.2	F9-39	Intelsat 35e	6.761	KC 39A	X	GTO
B1038	08/24/17	v1.2	F9-40	Formosat 5	0.475	VA 4E	DRL	SSO
B1038.2	02/22/18	v1.2	F9-50	Paz/Microsat 2a/b	~2.0	VA 4E	X	LEO
B1039	08/14/17	v1.2	F9-41	Dragon 13/CRS-12	~8.4	KC 39A	LZ1	ISS
B1039.2	04/02/18	v1.2	F9-53	Dragon 10.2/CRS-14	~8.125	CC 40	X	ISS
B1040	09/07/17	v1.2	F9-42	OTV-5 (X-37B)	~5.00	KC 39A	LZ1	LEO
B1040.2	06/04/18	v1.2	F9-57	SES 12	5.384	CC 40	X	GTO
B1041	10/09/17	v1.2	F9-44	Iridium Next 3	8.6	VA 4E	DRL	SSO
B1041.2	03/30/18	v1.2	F9-52	Iridium Next 5	8.6	VA 4E	X	LEO
B1042	10/30/17	v1.2	F9-45	Koreasat 5A	3.7	KC 39A	DRL	GTO
B1043	01/08/18	v1.2	F9-46	Zuma		CC 40	LZ1	LEO
B1043.2	05/22/18	v1.2	F9-56	Iridium 6/GRACE-FO	~5.5	VA 4E	X	LEO
B1044	03/06/18	v1.2	F9-51	Hispasat 1F	6.092	CC 40	X	GTO
B1045	04/18/18	v1.2	F9-54	TESS	0.35	CC 40	DRL	EEO
B1045.2	06/29/18	v1.2	F9-58	Dragon 11.2/CRS-15	~8.175	CC 40	X	ISS
B1046	05/11/18	v1.2	F9-55	Bangabandhu 1	~3.72	KC 39A	DRL	GTO
B1046.2	08/07/18	v1.2	F9-61	Merah Putih	5.8	CC 40	DRL	GTO-
B1046.3	12/03/18	v1.2	F9-65	Spaceflight SSO-A	~4.0	VA 4E	DRL	SSO
B1046.4	01/19/20	v1.2	F9-80	IFA Crew Dragon 2	~12.5	KC 39A	X	<SUB>
B1047	07/22/18	v1.2	F9-59	Telstar 19V	7.075	CC 40	DRL	GTO-
B1047.2	11/15/18	v1.2	F9-64	Es'hail 2	~5.3	KC 39A	DRL	GTO
B1047.3	08/06/19	v1.2	F9-75	AMOS 17	6.5	CC 40	X	GTO
B1048	07/25/18	v1.2	F9-60	Iridium NEXT 7	8.6	VA 4E	DRL	LEO
B1048.2	10/08/18	v1.2	F9-63	SAOCOM 1A	3.0	VA 4E	LZ4	SSO
B1048.3	02/22/19	v1.2	F9-70	Nusantara Satu+usats	4.85	CC 40	DRL	GTO+
B1048.4	11/11/19	v1.2	F9-76	Starlink 1	15.60	CC 40	DRL	LEO
B1048.5	03/18/20	v1.2	F9-84	Starlink 1 F5	15.6	KC 39A	DRL-X	LEO
B1049	09/10/18	v1.2	F9-62	Telstar 18V	7.06	CC 40	DRL	GTO-
B1049.2	01/11/19	v1.2	F9-68	Iridium NEXT 8	8.6	VA 4E	DRL	LEO
B1049.3	05/24/19	v1.2	F9-72	Starlink 0.9 (60 ea)	13.609	CC 40	DRL	LEO
B1049.4	01/07/20	v1.2	F9-79	Starlink 1 F2 (60ea)	15.6	CC 40	DRL	LEO
B1049.5	06/04/20	v1.2	F9-86	Starlink 1 F7 (60ea)	15.6	CC 40	DRL-J	LEO
B1049.6	08/18/20	v1.2	F9-92	Starlink 1-10 (58)	14.44	CC 40	DRL	LEO
B1049.7	11/25/20	v1.2	F9-101	Starlink 1-15 (60ea)	15.6	CC 40	DRL	LEO
B1049.8	03/04/21	v1.2	F9-108	Starlink 1-17 (60ea)	15.6	KC 39A	DRL	LEO
B1049.9	05/04/21	v1.2	F9-117	Starlink 1-25 (60ea)	15.6	KC 39A	DRL	LEO
B1049.10	09/14/21	v1.2	F9-126	Starlink 2-1 (51ea)	13.26	VA 4E	DRL	LEO
B1050	12/05/18	v1.2	F9-66	CRS-16/Dragon 12.2	~8.039	CC 40	LZ1-X	ISS
B1051	03/02/19	v1.2	F9-69	DM-1/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1051.2	06/12/19	v1.2	F9-73	RCM	4.29	VA 4E	LZ4	SSO
B1051.3	01/29/20	v1.2	F9-81	Starlink 1 F3 (60ea)	15.6	CC 40	DRL	LEO
B1051.4	04/22/20	v1.2	F9-85	Starlink 1 F6 (60ea)	15.6	KC 39A	DRL	LEO
B1051.5	08/07/20	v1.2	F9-89	Starlink 1 F9 (57ea)	15.192	KC 39A	DRL	LEO
B1051.6	10/18/20	v1.2	F9-97	Starlink 1-13 (60)	15.6	KC 39A	DRL	LEO
B1051.7	12/13/20	v1.2	F9-103	SXM-7	~7.0	CC 40	DRL	GTO-
B1051.8	01/20/21	v1.2	F9-106	Starlink 1-16 (60)	15.6	KC 39A	DRL	LEO
B1051.9	03/14/21	v1.2	F9-112	Starlink 1-21 (60)	15.6	KC 39A	DRL	LEO

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B1051.10	05/09/21	v1.2	F9-118	Starlink 1-27 (60)	15.6	CC 40	DRL	LEO
B1051.11	12/18/21	v1.2	F9-133	Starlink 4-4 (52)	~14.0?	VA 4E	DRL	LEO
B1051.12	03/19/22	v1.2	F9-145	Starlink 4-12 (53)	16.25	CC 40	DRL	LEO
B1051.13	07/17/22	v1.2	F9-166	Starlink 4-22 (53)	~15.9	CC 40	DRL	LEO
B1052	04/11/19	FH	FH-2	Arabsat 6A side		KC 39A	LZ-1/2	GTO+
B1052.2	06/25/19	FH	FH-3	STP-2 side		KC 39A	LZ-1/2	EEO
B1052.3	01/31/22	v1.2	F9-139	CSG FM2	2.2	CC 40	LZ-1	SSO
B1052.4	03/09/22	v1.2	F9-145	Starlink 4-10	~13.68	CC 40	DRL	LEO
B1052.5	05/18/22	v1.2	F9-156	Starlink 4-18	~15.9	KC 39A	DRL	LEO
B1052.6	08/04/22	v1.2	F9-169	KPLO	0.678	CC 40	DRL	EEO
B1053	04/11/19	FH	FH-2	Arabsat 6A side		KC 39A	LZ-1/2	GTO+
B1053.2	06/25/19	FH	FH-3	STP-2 side		KC 39A	LZ-1/2	EEO
B1054	12/23/18	v1.2	F9-67	GPS 3-1	4.4	CC 40	X	MTO
B1055	04/11/19	FH	FH-2	Arabsat 6A core	6.465	KC 39A	DRL-X	GTO+
B1056	05/04/19	v1.2	F9-7 1	CRS-17/Dragon 13.2	~8.0	CC 40	DRL	ISS
B1056.2	07/25/19	v1.2	F9-74	CRS-18/Dragon 8.3	~7.831	CC 40	LZ1	ISS
B1056.3	12/17/19	v1.2	F9-78	JCSat 18/Kacific 1	6.956	CC 40	DRL	GTO-
B1056.4	02/17/20	v1.2	F9-82	Starlink 1 F4	15.6	CC 40	DRL-X	LEO
B1057	06/25/19	FH	FH-3	STP-2 core	~2.94	KC 39A	DRL-X	EEO
B1058	05/30/20	v1.2	F9-87	DM-2 Crew Dragon	~12.5	KC 39A	DRL	ISS
B1058.2	07/20/20	v1.2	F9-91	ANASIS 2		CC 40	DRL	GTO+
B1058.3	10/06/20	v1.2	F9-95	Starlink 1-12	15.6	KC 39A	DRL	LEO
B1058.4	12/06/20	v1.2	F9-102	CRS-21/Dragon 208.1	~12.5	KC 39A	DRL	ISS
B1058.5	01/24/21	v1.2	F9-107	Transporter 1		CC 40	DRL	SSO
B1058.6	03/11/21	v1.2	F9-111	Starlink 1-20	15.6	CC 40	DRL	LEO
B1058.7	04/07/21	v1.2	F9-114	Starlink 1-23	15.6	CC 40	DRL	LEO
B1058.8	05/15/21	v1.2	F9-119	Starlink 1-26	~13.7	KC 39A	DRL	LEO
B1058.9	11/13/21	v1.2	F9-129	Starlink 4-1	~15.5?	CC 40	DRL	LEO
B1058.10	01/13/22	v1.2	F9-137	Transporter 3	~2.5	CC 40	LZ1	SSO
B1058.11	02/21/22	v1.2	F9-142	Starlink 4-8	~13.1	CC 40	DRL	LEO
B1058.12	05/06/22	v1.2	F9-153	Starlink 4-17	~15.9	KC 39A	DRL	LEO
B1058.13	07/07/22	v1.2	F9-163	Starlink 4-21	~15.9	CC 40	DRL	LEO
B1059	12/05/19	v1.2	F9-77	CRS-19/Dragon 6.3	~7.804	CC 40	DRL	ISS
B1059.2	03/07/20	v1.2	F9-83	CRS-20/Dragon 12.3	~7.259	CC 40	LZ1	ISS
B1059.3	06/13/20	v1.2	F9-88	Starlink 1 F8+	15.41	CC 40	DRL	LEO
B1059.4	08/30/20	v1.2	F9-93	SAOCOM 1B	3.05	CC 40	LZ1	SSO
B1059.5	12/19/20	v1.2	F9-104	NROL-108		KC 39A	LZ1	LEO
B1059.6	02/16/21	v1.2	F9-110	Starlink 1-19	15.6	CC 40	DRL-X	LEO
B1060	06/30/20	v1.2	F9-90	GPS 3-3	4.311	CC 40	DRL	MTO
B1060.2	09/03/20	v1.2	F9-94	Starlink 1-11	15.6	KC 39A	DRL	LEO
B1060.3	10/24/20	v1.2	F9-98	Starlink 1-14	15.6	CC 40	DRL	LEO
B1060.4	01/08/21	v1.2	F9-105	Turksat 5A	3.5	CC 40	DRL	GTO+
B1060.5	02/04/21	v1.2	F9-109	Starlink 1-18	15.6	CC 40	DRL	LEO
B1060.6	03/24/21	v1.2	F9-113	Starlink 1-22	15.6	CC 40	DRL	LEO
B1060.7	04/29/21	v1.2	F9-116	Starlink 1-24	15.6	CC 40	DRL	LEO
B1060.8	06/30/21	v1.2	F9-124	Transporter 2	~2.4	CC 40	LZ1	SSO
B1060.9	12/02/21	v1.2	F9-131	Starlink 4-3	~14.0?	CC 40	DRL	LEO
B1060.10	01/19/22	v1.2	F9-128	Starlink 4-6	~14.0?	KC 39A	DRL	LEO
B1060.11	03/03/22	v1.2	F9-144	Starlink 4-9	~13.4?	KC 39A	DRL	LEO
B1060.12	04/21/22	v1.2	F9-150	Starlink 4-14	~15.9	CC 40	DRL	LEO
B1060.13	06/17/22	v1.2	F9-157	Starlink 4-19	~15.9	KC 39A	DRL	LEO
B1061	11/16/20	v1.2	F9-99	Crew 1/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1061.2	04/23/21	v1.2	F9-115	Crew 2/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1061.3	06/06/21	v1.2	F9-122	SXM-8	~7.0	CC 40	DRL	GTO-
B1061.4	08/29/21	v1.2	F9-125	CRS-23/Cargo Dragon	~12.5	KC 39A	DRL	ISS
B1061.5	12/09/21	v1.2	F9-132	IXPE	0.337	KC 39A	DRL	LEO
B1061.6	02/03/22	v1.2	F9-141	Starlink 4-7	~14.0	KC 39A	DRL	LEO

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B1061.7	04/01/22	v1.2	F9-147	Transporter 4		CC 40	DRL	SSO
B1061.8	05/25/22	v1.2	F9-157	Transporter 5	~2.0	CC 40	LZ1	SSO
B1061.9	06/19/22	v1.2	F9-161	Globalstar FM15	~0.7	CC 40	DRL	LEO
B1061.10	08/12/22	v1.2	F9-171	Starlink 3-3	~12.9	VA 4E	DRL	SSO
B1062	11/05/20	v1.2	F9-96	GPS 3-4	4.311	CC 40	DRL	MTO
B1062.2	06/17/21	v1.2	F9-123	GPS 3-5	4.311	CC 40	DRL	MTO
B1062.3	09/16/21	v1.2	F9-127	Inspiration 4	~12.5	KC 39A	DRL	LEO
B1062.4	01/06/22	v1.2	F9-136	Starlink 4-5	~14.0	KC 39A	DRL	LEO
B1062.5	04/08/22	v1.2	F9-148	Axiom 1/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1062.6	04/29/22	v1.2	F9-152	Starlink 4-16	~15.9	CC 40	DRL	LEO
B1062.7	06/08/22	v1.2	F9-158	Nilesat 301	3.938	CC 40	DRL	GTO+
B1062.8	07/24/22	v1.2	F9-168	Starlink 4-25	~15.9	KC 39A	DRL	LEO
B1062.9	08/19/22	v1.2	F9-172	Starlink 4-27	~15.9	CC 40	DRL	LEO
B1063	11/21/20	v1.2	F9-100	Sentinel 6	1.192	VA 4E	LZ4	LEO
B1063.2	05/26/21	v1.2	F9-120	Starlink 1-28	15.6	CC 40	DRL	LEO
B1063.3	11/24/21	v1.2	F9-130	DART	0.61	VA 4E	DRL	HCO
B1063.4	02/25/22	v1.2	F9-143	Starlink 4-11	~14.2	VA 4E	DRL	LEO
B1063.5	05/13/22	v1.2	F9-154	Starlink 4-13	~15.9	VA 4E	DRL	LEO
B1063.6	07/11/22	v1.2	F9-164	Starlink 3-1	~13.8	VA 4E	DRL	LEO
B1063.7	08/31/22	v1.2	F9-174	Starlink 3-4	~13.8	VA 4E	DRL	SSO
B1067.1	06/03/21	v1.2	F9-121	CRS-22/Cargo Dragon	~12.5	KC 39A	DRL	ISS
B1067.2	11/11/21	v1.2	F9-128	Crew 3/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1067.3	12/19/21	v1.2	F9-134	Turksat 5B	4.5	CC 40	DRL	GTO+
B1067.4	04/27/22	v1.2	F9-151	Crew 4/Crew Dragon	~12.5	KC 39A	DRL	ISS
B1067.5	07/15/22	v1.2	F9-165	CRS-25/Cargo Dragon	~13.5	KC 39A	DRL	ISS
B1069.1	12/21/21	v1.2	F9-135	CRS-24/Cargo Dragon	~13.5	KC 39A	DRL-X+	ISS
B1069.2	08/28/22	v1.2	F9-173	Starlink 4-23	16.7	CC 40	DRK	LEO
B1071.1	02/02/22	v1.2	F9-140	NROL-87		VA 4E	LZ4	SSO
B1071.2	04/17/22	v1.2	F9-149	NROL-85 (2xIntruder)	~6.5	VA 4E	LZ4	LEO
B1071.3	06/18/22	v1.2	F9-160	SARah 1	~4.0	VA 4E	LZ4	SSO
B1071.4	07/22/22	v1.2	F9-167	Starlink 3-2	~13.8	VA 4E	DRL	SSO
B1073.1	05/14/22	v1.2	F9-155	Starlink 4-15	~15.9	CC 40	DRL	LEO
B1073.2	06/29/22	v1.2	F9-162	SES 22	3.5	CC 40	DRL	GTO
B1073.3	08/10/22	v1.2	F9-170	Starlink 4-26	~15.6	KC 39A	DRL	LEO

Estimated v1.2 First Stage Block Number Ranges

(Second Stage Number Ranges Differ)

 Block 1 B1019-B1020
 Block 2 B1021-B1027
 Block 3 B1028-B1038
 Block 4 B1039-B1045
 Block 5 B1046-

First Stage Landing Stats
 as of 08/31/2022

 Successes/Attempts 138/150*+
 F9 Successes/Attempts 132/141+
 FH Successes/Attempts 6/9*
 OCISLY-CC/KC 44/54*
 OCISLY-VA 9/9
 JRTI-VA 7/8
 JRTI-CC/KC 29/29+
 ASOG-CC/KC 20/20
 CC LZ-1 20/21
 CC LZ-2 3/3
 VA LZ-4 6/6

* FH-2 Core B1055.1 landed successfully on OSCILY, but subsequently
 toppled and was lost. Is not counted as a success in these numbers.

+ F9-136 first stage B1069.1 landed successfully on JRTI but was
 damaged during post-landing securement. It is counted as a
 success in these numbers because plans called for its repair
 to fly again. It did fly again after complete engine replacement
 and other repairs.